Technische Universität München

Mathematics preparatory course

2026

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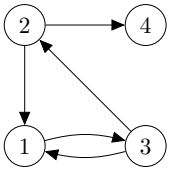
Blatt 3

4 Relations

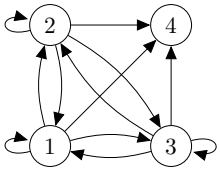
4.1 Properties of relations

For every of the depicted relations on $\{1, 2, 3, 4\}$ decide whether it is reflexive, irreflexive, symmetric, antisymmetric, asymmetric, and transitive, respectively?

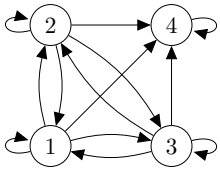
(a)



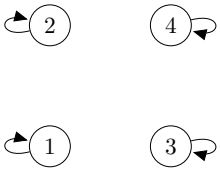
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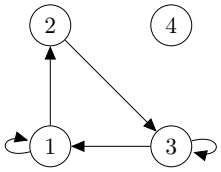
(c)



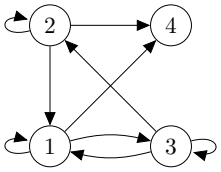
(d)



(e)



(f)



4.2 Properties of relations (2)

Consider the set

$$M = \{x, y, z\}$$

For each subproblem, define a binary relation on M satisfying the stated properties:

(a) reflexive, but not symmetric

(b) neither symmetric nor antisymmetric

(c) antisymmetric, but not asymmetric

(d) not transitiv, but each pair of elements is in relation with each other in at least one of the two possible ways

(e) symmetric and each pair of elements is in relation with each other in at least one of the two possible ways

(f) reflexive and asymmetric

4.3 Partial orders

Consider the sets

$$M = \{x, y\}, \quad N = \{x, y, z\}$$

(a) Graphically depict all partial orders on the set M.

(b) Which of these can be turned into each other simply by renaming the elements of M?

Formally, this is called "isomorphic" where "isos" is Greek for "equal" and "morphe" is again Greek for "structure, form".

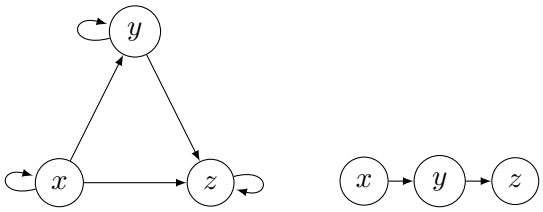
(c) Graphically depict all partial orders on the set N "up to isomorphim" (also: "modulo isomorphism"), i.e. again only list the graphs that remain if you do not care for the names of the nodes.

Remark:

As partial orders are required to be reflexive and transitive, a typical convention is to not explicitly draw self-loops and edges that arise from transitiviy. E.g. the relation

$$R = \{(x, x), (x, y), (x, z), (y, y), (y, z), (z, z)\}$$

can be simplified to just drawing the edges (x, y) and (y, z), since (x, x), (y, y) and (z, z) are reflexive and (x, z) is transitively derived from (x, y) and (y, z).



(d) (*) Wrt. a partial order R, an element x is a *maximal element* (resp. *minimal element*) wrt. R if (def) there is no element y with $x \neq y$, so that xRy (resp. yRx).

Graphically this means that an maximal (resp. minimal) element does not point at any element other than itself (resp. no *other* element points at a minimal element) (just for clarification: self-loops are still allowed resp. required).

N.B.: "maximal element" is not the same as "maximum", see (f).

For each of the partial orders of (c), determine there maximal and minimal elements.

(e) (*) For comparison: if R is partial order, then an element x is called the *greatest element* (resp. *least element*) wrt. R if (def) for every element y we have yRx (resp. xRy).

Graphically this means that every (all) element points at **the** greatest, resp. **the** least element point at every element (including the self-loop).

In particular, there can only be at most one greatest resp. least element, while there might be several maximal resp. minimal elements.

Again, decide for the partial orders of (c) which of them have a greatest resp. a least element.

(f) (*) The "maximum" is the greatest element in a subset $S \subseteq \Omega$ of some (Ω, R) partially ordered set (i.e. Ω is a set, $R \subseteq \Omega \times \Omega$ is a partial order on Ω.)

For each relation determined in (c) decide which subsets of M have a maximum.

(g) (**) Show: a partial order on a finite set has always a maximal and also a minimal element (use induction).

Also given an example of a partial order which does not have a maximal element.

(h) (**) Define a partial order which has exactly one maximal element and yet no greatest element.

(Hint: $\mathbb{N}_0$ wrt. $\leq_{\mathbb{N}_0}$ does not have a greatest element. Add a maximal element!)

4.4 Equivalence relations

(a) Graphically depict all equivalence relations on the set $\{1, 2, 3\}$.

(b) The *equivalence class* of an element a wrt. a given equivalence relation $\sim \subseteq A \times A$ is the set of all elements that are equivalent to a. Often, this set is denoted by $[a]_\sim := \{a' \in A \mid a \sim a'\}$.

Graphically this means that $[a]_\sim$ is simply the set of all elements that point at a resp. that are pointing at a (as an equivalence relation is symmetric, this is the same; and as an equivalence relation is transitive, if a points at b, and b points at c, then a also points at c, and thus also c at a).

For each of the equivalence relations of (a), determine all equivalence classes. (Note that there is at least one, but at most $|A|$ many equivalence classes.)

(c) (*) In (a) you have already seen that wrt. $n = 3$ elements, there are exactly 3 equivalence relations which have exactly two equivalence classes.

Determine the number of equivalence relations on $\{1, 2, 3, 4\}$ with exactly 2 equivalence classes.

Try to understand how the problem can be reduced to counting the equivalence relations on $\{1, 2, 3\}$ which have either 1 or 2 equivalence classes.

Try to also determine the number of equivalence relations on $\{1, 2, 3, 4, 5\}$ with exactly 2 equivalence classes.

Can you inductively define how to compute the number of equivalence relations with exactly two equivalence classes on $\{1, 2, \ldots, n\}$?

(These numbers are usually denoted by $S_{n,2}$.)

4.5 Modulo

For a fixed $n \in \mathbb{N}$ the equivalence relation $\equiv_n$ is defined on $\mathbb{Z}$ by:

$$\text{We write } a \equiv_n b \text{ IF: } \tfrac{a-b}{n} \in \mathbb{Z} \text{ and otherwise } a \not\equiv_n b.$$

$a \equiv_n b$ can also be read as "a and b have the same remainder ("residue") when divided by n".

The equivalence class of a number $a \in \mathbb{Z}$ is denoted by $[a]_n = \{b \in \mathbb{Z} \mid a \equiv_n b\}$.

(The relation $\equiv_n$ was studied already by Gauss in his disquisitiones arithmeticae; it is often refered to also as "congruence (relation)" and plays an important role in number theory with applications in cryptography.)

(a) Determine the equivalences classes of $\equiv_n$ for $n \in \{3, 5, 15\}$.

E.g. first consider the set $\{0, 1, 2 \ldots, 44\}$ and determine which numbers are equivalent.

(b) (*) Let $m, n \in \mathbb{N}$ be any two positive integers. Show:

- If m divides n and $a \equiv_n b$, then also $a \equiv_m b$; hence: $[a]_n \subseteq [a]_m$.

  Recall that "m divides n (without remainder)" is usually denoted by "$m|n$" and "$|$" is a partial order on $\mathbb{N}$.
- $[a]_m \cap [a]_n = [a]_k$ for k the least common multiple of m and n.
- $[a]_m \cap [a]_n = [a]_{mn}$ IFF the greatest common divisor of m and n is 1.

5 Functions

5.1 Neither Injective nor Surjective

Define a function that is neither injective nor surjective.

5.2 Injectivity and surjectivity

Check, whether the following functions are injective and/or surjective:

(a) $f\colon \mathbb{Z} \to \mathbb{N}_0,\ x \mapsto x^2$

(b) $f\colon \mathbb{N}_0 \to \mathbb{N}_0,\ x \mapsto x^2$

(c) $f\colon \mathbb{N} \to \mathbb{N},\ x \mapsto \begin{cases} 1 & \text{if } x = 1 \\ x - 1 & \text{else} \end{cases}$

(d) $f\colon \mathbb{Z} \to \mathbb{Z},\ x \mapsto x - 1$

5.3 Graphical representation of functions

Let $A = \{1, 2, 3\}$.

1. How many different functions $f : A \to A$ are there? How many of them are bijective?
2. Consider the two functions $f, g: A \to A$ defined as follows:

$$f(1) = 2, \quad f(2) = 3, \quad f(3) = 1 \qquad g(1) = 2, \quad g(2) = 1, \quad g(3) = 3$$

   (a) Depict f and g graphically both as (i) a bipartite graph with separate nodes for the pre-images (left) and images (right), and (ii) also as relation on A with only one node for each element of A.

   (b) As in (a), also visualize the following functions:

$$f \circ g, \quad g \circ f, \quad f \circ f, \quad f \circ f \circ f, \quad g \circ g, \quad g \circ g \circ g$$

   (c) (*) Show: for every bijective self-map (permutation) $h: A \to A$ we always have that for every $a \in A$:

$$(h \circ h \circ h \circ h \circ h \circ h)(a) = a$$

5.4 Properties of functions (*)

Argue why the following statements hold for arbitrary (apart from the explicit restrictions in the statement itself) functions f and g.

1. If $f\colon A \to B$ is injective, then $g\colon A \to f(A),\ \ a \mapsto f(a)$ is bijective.
2. If $f\colon A \to A$ is injective and A is finite, then f is bijective.
3. If $f\colon A \to A$ is surjective and A is finite, then f is bijective.
4. If $f\colon A \to B$ and $g\colon B \to C$ are both injective (surjective), then so is $(g \circ f)$.
5. Let $f\colon A \to B$ and $g\colon B \to C$. If $(g \circ f)$ is surjective, then so is g.
6. Let $f\colon A \to B$ and $g\colon B \to C$. If $(g \circ f)$ is injective, then so is f.

5.5 Groups*

A function $f: A \times A \to A$ is also called a binary operation on A.

In case of a binary operation, we usually write (afa) for $f(a, a)$ as e.g. for addition and multiplication.

A set G together with a binary operation $\odot : G \times G \to G$ is called a group, if the following conditions are all satisfied:

Associativity:	for all $a, b, c \in G : a \odot (b \odot c) = (a \odot b) \odot c$
Neutral Element:	exists $n \in G :$ for all $a \in G : n \odot a = a = a \odot n$
Inverse Elements:	for all $a \in G :$ exists $b \in G : a \odot b = n = b \odot a$, whereby n is the neutral element defined in the previous condition

1. Show: For any set A the set defined by $\mathcal{S}_A := \{f : A \to A \mid f \text{ is bijective}\}$ is a group w.r.t. the composition operator $\circ$.

   Remark: This group $\mathcal{S}_A$ is commonly being referred to as the *symmetric group* on/over A. It is related to the study of symmetries of e.g. crystals (i.e. graph automorphism). In some sense it is the most general group, and thus plays an important role everywhere in mathematics and computer science (e.g. sorting).

2. Show, only using the definitions of *associativity*, the *neutral element* and the *inverse* of an element, that
   (a) each group has exactly one neutral element, i.e. if both $n \in G$ and $n' \in G$ satisfy the definition of a neutral element, then $n = n'$.
   (b) each element $a \in G$ has exactly one inverse, i.e. if both $b \in G$ and $b' \in G$ satisfy the definition if an inverse of a, then $b = b'$.
   (For this reason, the unique inverse of an element $a \in G$ in a group is usually denoted by a^{-1}.)

5.6 Cantor's Pairing Function*

The function

$$f\colon \mathbb{N}\times\mathbb{N}\to\mathbb{N},\ (m,n)\mapsto \frac{1}{2}(m+n-1)(m+n-2)+m$$

is called "Cantor's paring function. It can be shown to be a bijection (see the following subproblems) which yields the formal argument why $\mathbb{N}$ and $\mathbb{N}\times\mathbb{N}$ and thus also $\mathbb{Q}$ have all the same cardinality.

(a) First, complete the following table to get a basic idea of the function:

(Feel tree to use a computer.)

n \ m	1	2	3	4	5
1					
2					
3					
4					
5					

(b) Show that f can indeed only return values in $\mathbb{N}$.

(c) Let $n\in\mathbb{N}$ be fixed and consider the values $f(m,n+1-m)$ for $m=1,\dots,n$ on the n-th diagonal $f(1,n),f(2,n-1),\dots,f(n,1)$.

Simplify the definition of $f(m,n+1-m)$ under the assumption that n is fixed and $1\le m\le n$.

Then compute both $f(m+1,n+1-(m+1))-f(m,n+1-m)$ (assuming that $m<n$), and also $f(1,n+1)-f(n,1)$.

(d) (*) Show, that f is surjective.

(e) (*) Show, that f is injective.

Hint: Assume, that for arbitrary, but fixed, $m,n,r,s\in\mathbb{N}: f(m,n)=f(r,s)$ and then distinguish the two cases that either $m+n=r+s$ or $m+n\neq r+s$.